

2.6 - Proving Geometric Relationships

Theorems are statements we can prove — today we prove four of them.

Right Angles Congruence Theorem

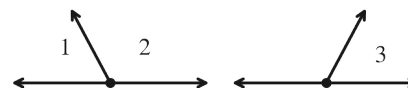
All right angles are _____.



Congruent Supplements & Complements Theorems

If two angles are supplementary to the same angle (or to congruent angles), then they are _____.

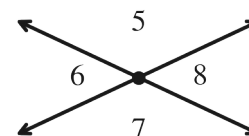
If two angles are complementary to the same angle (or to congruent angles), then they are _____.



Linear Pair Postulate & Vertical Angles Theorem

If two angles form a linear pair, then they are _____.

Vertical angles are _____.



Every proof in this section has the same shape. A geometry word goes in — right, perpendicular, supplementary, vertical — and a $\cong$ comes out. In between, it is numbers and 2.4 algebra. Unwrap, work, wrap.

THE 2.6 MOVE — unwrap, do the algebra, wrap it back up

1. UNWRAP the geometry word into a number sentence. (Definition of right angle / supplementary / congruent angles)
2. Do the algebra. This is 2.4. (Substitution · Reflexive · Subtraction · Transitive)
3. WRAP the number sentence back into $\cong$. (Definition of congruent angles)
4. Once a theorem has been proved, it becomes a ONE-LINE reason. Example 4 uses Example 3 in a single line.

Part 1 — The dictionary: what each word unwraps to

Line 2 of every proof today is one of these. The last row is the WRAP — it runs the other way.

The statement says	So you may write	Reason
$\angle 1$ is a right angle		
Segment $AB \perp$ segment BC		
$\angle 1$ and $\angle 2$ are supplementary		
$\angle 1$ and $\angle 2$ are complementary		
$\angle 1$ and $\angle 2$ form a linear pair		
$m\angle 1 = m\angle 2$ (the WRAP)		

Part 2 — Example 1: Prove the Right Angles Congruence Theorem

The statements are done. Supply each reason. Watch the shape: unwrap on line 2, algebra on line 3, wrap on line 4.

Given: $\angle 1$ and $\angle 2$ are right angles. Prove: $\angle 1 \cong \angle 2$.

Statements	Reasons
1. $\angle 1$ and $\angle 2$ are right angles	1.
2. $m\angle 1 = 90^\circ$; $m\angle 2 = 90^\circ$	2.
3. $m\angle 1 = m\angle 2$	3.
4. $\angle 1 \cong \angle 2$	4.

REASON BANK

Given · Definition of right angle · Definition of perpendicular · Definition of supplementary angles

Definition of complementary angles · Definition of congruent angles · Definition of linear pair (from the diagram)

Linear Pair Postulate · Right Angles Congruence Theorem · Congruent Supplements Theorem

Substitution Property of Equality · Reflexive Property of Equality · Subtraction Property of Equality · Transitive Property of Equality

Part 3 — Example 2: Perpendicular segments

The reasons are done. Write each statement. Line 3 is the first time a theorem YOU PROVED is a one-line reason.



Given: Segment $PQ \perp$ segment QR , and segment $SR \perp$ segment QR . Prove: $\angle Q \cong \angle R$.

Statements	Reasons
1.	1. Given
2.	2. Definition of perpendicular
3.	3. Right Angles Congruence Theorem

Part 4 — Example 3: Prove the Congruent Supplements Theorem

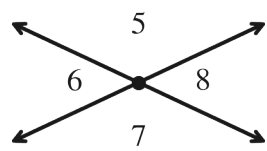
Note: You have done this proof before. It is the Overlap Move: two sums that share $m\angle 2$, and you cover the shared piece. The only new thing is the unwrap on line 2 and the wrap on line 6.

Given: $\angle 1$ and $\angle 2$ are supplementary; $\angle 2$ and $\angle 3$ are supplementary. Prove: $\angle 1 \cong \angle 3$.

Statements	Reasons
1.	1. Given
2. $m\angle 1 + m\angle 2 = 180^\circ$; $m\angle 2 + m\angle 3 = 180^\circ$	2.
3.	3. Substitution Property of Equality
4. $m\angle 2 = m\angle 2$	4. Reflexive Property of Equality
5. $m\angle 1 = m\angle 3$	5.
6.	6. Definition of congruent angles

Part 5 — Example 4: Prove the Vertical Angles Theorem

Nothing is filled in. Two hints: line 2 comes from the DIAGRAM, and line 4 is all of Example 3 in one line.



Given: $\angle 5$ and $\angle 7$ are vertical angles. Prove: $\angle 5 \cong \angle 7$.

Statements	Reasons
1.	1.
2.	2.
3.	3.
4.	4.

Part 6 — Example 5: Geometry and Algebra

Now the theorems do the work for you. Name the one you are using each time.

a) In the top diagram, $m\angle 1 = 123^\circ$. Find $m\angle 2$, $m\angle 3$, and $m\angle 4$.		
b) In the bottom diagram, find the value of w .		

Part 7 — On your own: Prove the Congruent Complements Theorem

Same shape as Example 3 with 90° in place of 180° . No bank.

Note: Do NOT write “Congruent Complements Theorem” as a reason — that is the statement you are proving. Using it would be circular.

Given: $\angle 1$ and $\angle 2$ are complementary; $\angle 2$ and $\angle 3$ are complementary. Prove: $\angle 1 \cong \angle 3$.

Statements	Reasons
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.

BEFORE YOU HAND THIS IN

Line 1 is the given, word for word. The last line is exactly what you were asked to prove.
Line 2 UNWRAPS a word into numbers. The last line WRAPS numbers back into $\cong$. Everything between is algebra.
A theorem is a reason only AFTER it has been proved. Never cite the theorem you are proving.
If a line uses a fact, that fact is written on an earlier line.